Software Engineering Lab 7
Fuzzing for Input Validation Bugs

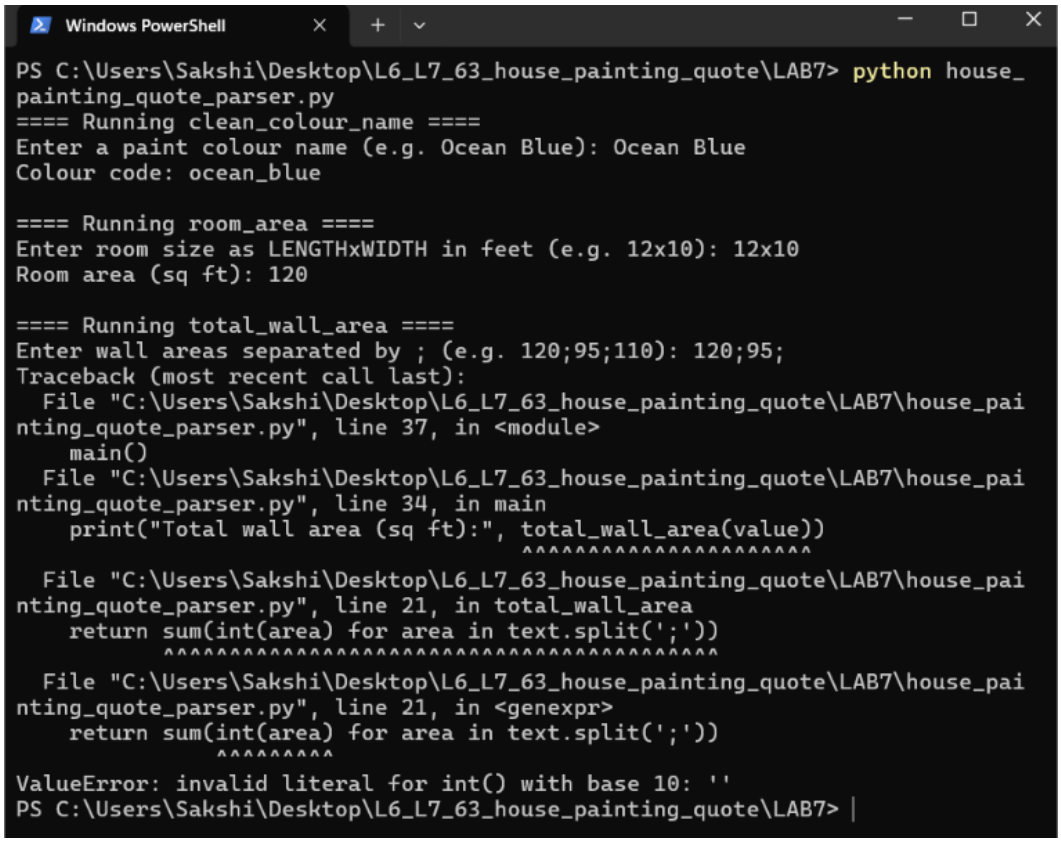
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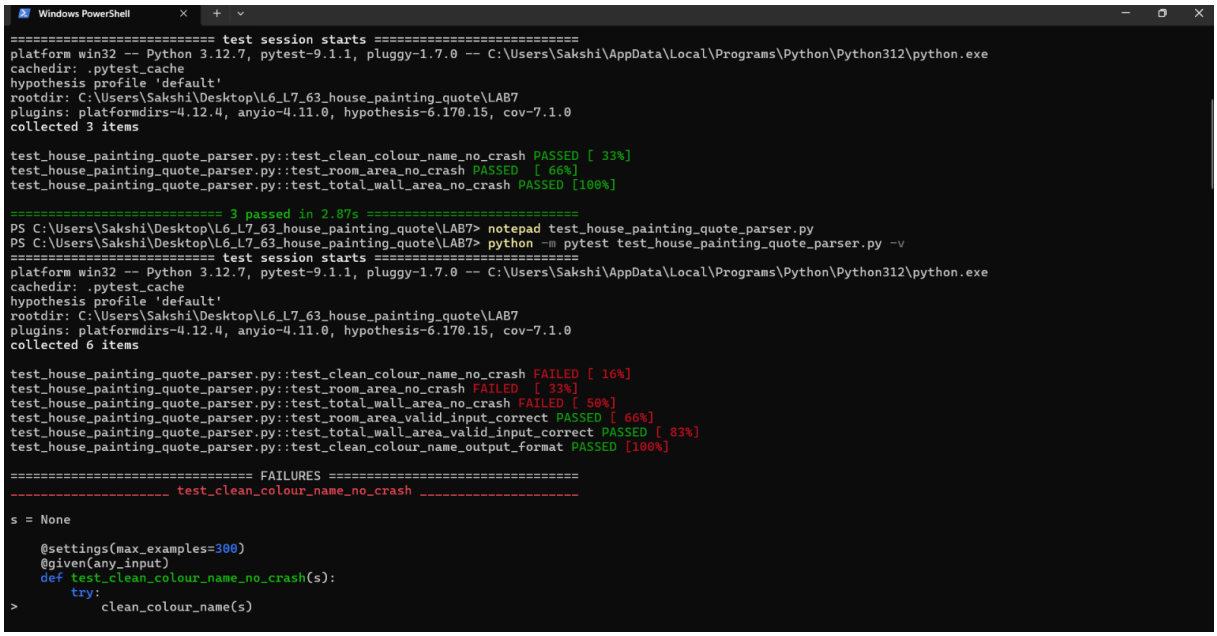
Repo: L6_L7_63_house_painting_quote

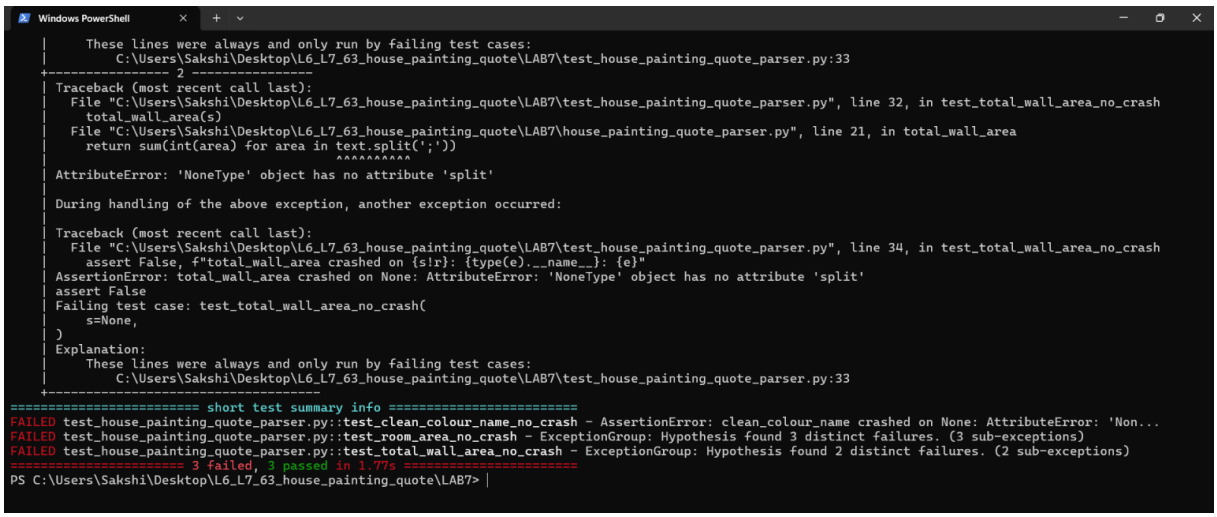
Step 2: Running the buggy program (SS1)



The first two functions gave output but total_wall_area crashed when I entered 120;95; with a semicolon at the end.

Step 4: Running the tests on the buggy program (SS2)





3 tests failed and 3 passed. The 3 no crash tests failed for all three functions. The other 3 tests only give valid inputs so they passed.

Inputs found by Hypothesis that crashed the functions:

Function	Input	Error
clean_colour_name	None	AttributeError
room_area	'X'	ValueError (int of empty string)
room_area	'' (empty)	ValueError (not enough values to unpack)
room_area	None	AttributeError
total_wall_area	'' or 120;95;	ValueError (int of empty string)
total_wall_area	None	AttributeError

Step 5: Fixing the program (SS3)

Changes made in house_painting_quote_parser.py:

1. Added isinstance check in all functions so None or non string input returns None.

2. In room_area, checked that there are exactly two parts after splitting on x and both are numbers greater than 0.

3. In total_wall_area, empty entries are skipped, so 120;95; gives 215.

4. Numbers are accepted only if they are digits 0 to 9, so inputs like -5x10 are rejected.

5. main() prints Invalid message if the function returns None.

```
PS C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7> notepad house
_painting_quote_parser.py
PS C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7> python house_
painting_quote_parser.py
==== Running clean_colour_name ====
Enter a paint colour name (e.g. Ocean Blue): Ocean Blue
Colour code: ocean_blue

==== Running room_area ====
Enter room size as LENGTHxWIDTH in feet (e.g. 12x10): 12x10
Room area (sq ft): 120

==== Running total_wall_area ====
Enter wall areas separated by ; (e.g. 120;95;110): 120;95;
Total wall area (sq ft): 215
PS C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7>
```

Same inputs as before, now there is no crash.

Step 6: Running the tests on the fixed program (SS4)

```
PS C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7> python -m pytest test_house_painting_quote_parser.py -v
========================= test session starts =========================
platform win32 -- Python 3.12.7, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\Sakshi\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7
plugins: platformdirs-4.12.4, anyio-4.11.0, hypothesis-6.170.15, cov-7.1.0
collected 6 items

test_house_painting_quote_parser.py::test_clean_colour_name_no_crash PASSED [ 16%]
test_house_painting_quote_parser.py::test_room_area_no_crash PASSED  [ 33%]
test_house_painting_quote_parser.py::test_total_wall_area_no_crash PASSED [ 50%]
test_house_painting_quote_parser.py::test_room_area_valid_input_correct PASSED [ 66%]
test_house_painting_quote_parser.py::test_total_wall_area_valid_input_correct PASSED [ 83%]
test_house_painting_quote_parser.py::test_clean_colour_name_output_format PASSED [100%]

========================== 6 passed in 1.04s ==========================
PS C:\Users\Sakshi\Desktop\L6_L7_63_house_painting_quote\LAB7>
```

All 6 tests passed.